

# Jessica Song

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## EDUCATION

### Carnegie Mellon University

Pittsburgh, PA

*B.S. in Computer Science and Statistics & Machine Learning*

*Graduating May 2029*

- **Relevant Coursework:** Principles of Imperative Computation, Fundamentals of Programming and CS, Matrices and Linear Transformations, Integration and Approximation, Differential and Integral Calculus, Reasoning with Data, Physics I for Engineering Students

## EXPERIENCE

### Software Developer

June 2026 – Present

*ScottyLabs*

*Pittsburgh, PA*

- Built CMU Housing, a dorm-comparison website used by ~5,000 CMU students
- Designed and built a reusable React/TypeScript “Share” dropdown supporting URL copy, SMS, email, and iframe embeds, expanding site distribution to 4 channels

## PROJECTS

### PackRight | *Python, FastAPI, React, TypeScript, Three.js, Gemini API*

September 2026

- Designed and built an end-to-end 3D suitcase-packing planner spanning GLB geometry ingestion, an AI enrichment layer, a two-phase deterministic optimizer, and an independent validator, across 21 backend modules and 20+ frontend files
- Integrated the Gemini API to infer item metadata from rendered previews under a strict JSON schema, while architecturally barring the AI from setting item dimensions, weight, or plan validity
- Built a two-phase solver (exact/beam-search selection + rotation-aware 3D placement) and an independent validator that re-derives bounds, collisions, and support from an immutable snapshot; verified with 95 passing tests (86 pytest, 9 Vitest)

### 3D Chess Bot | *Java*

May 2026 – June 2026

- Engineered a Java chess engine implementing iterative-deepening minimax search with alpha-beta pruning, reaching 5-7 ply depth within a 5-second move limit
- Optimized search performance with MVV-LVA move ordering, killer-move and history heuristics, and time-boxed iterative deepening to guarantee a legal move every turn
- Reverse-engineered the game engine’s bytecode to uncover undocumented win conditions and designed a custom evaluation function (material, board control, king safety) to exploit them

### Endless Strands | *JavaScript, jQuery, SEO*

August 2025 – Present

- Engineered and shipped the organization’s website, serving as its primary web presence for donor and volunteer acquisition
- Integrated Google Analytics (GA4) for site-wide event tracking and applied on-page SEO practices (XML sitemap, semantic HTML, meta descriptions), increasing organic search impressions by [X]% within [Y] months
- Built a GitHub Actions CI/CD pipeline that auto-deploys the site and blocks commits/PRs containing hardcoded credentials via detect-secrets scanning

## HONORS

**MIT/Jane Street Math Prize for Girls Qualifier** (Top 3 Canadian Score)

October 2025

**PhysicsBowl Division 2 Contestant** (Top 5 score in team, helped place 1st in Canada)

June 2026

**4x AIME Qualifier**

2022 – 2026

## TECHNICAL SKILLS

**Languages:** C/C++, Java, Python, TypeScript, JavaScript, HTML/CSS

**Frameworks:** React, FastAPI, Node.js, Wordpress

**Developer Tools:** Git, GitHub, GitHub Actions, GitHub Pages, Docker, Vite

**Libraries:** jQuery, NumPy, SciPy, Pydantic, Three.js, Matplotlib

**Web, Data & APIs:** Shopify, Google Analytics (GA4), SEO, Open Graph, SQLite, Gemini API